

Sharps of Universally Baire Sets

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The goal of this note is to present a proof of the following theorem.

Theorem 1 (Woodin). *Suppose*

1. *there exists a proper class of Woodin cardinals,*
2. *A is a universally Baire set.*

Then

- a. $(\mathbb{R}, A)^\sharp$ *is universally Baire,*
- b. *for every V -generic filter g , $((\mathbb{R}^V, A)^\sharp)_g = (\mathbb{R}_g, A_g)^\sharp$.*

Proof. A more precise fact is proved as Theorem 7.¹

□

Notation 2. Suppose

1. λ is a limit of Woodin cardinals,
2. A is a $<\lambda$ -universally Baire set,
3. h is V -generic for $\text{Col}(\omega, <\lambda)$.

Then A_h^* is defined as

$$\bigcup_{\alpha < \lambda} A_{h \cap \text{Col}(\omega, < \alpha)}.$$

□

Lemma 3. *Suppose*

1. $\mathbb{P}$ *is a poset,*
2. A *is a $\mathbb{P}$ -universally Baire set,*
3. g *is V -generic for $\mathbb{P}$,*
4. M *is an inner model of $V[g]$ containing all reals,*

¹Compare with the proof of [Lar04, Theorem 3.3.9].

5. $j : V \longrightarrow M$ is elementary.

Then $j(A) = A_g$.

Proof. This is what the proof of [Kas25, Lemma 6.6.2] actually shows. $\square$

Proposition 4. *Suppose*

1. λ is a limit of Woodin cardinals,
2. $\kappa > \lambda$ is measurable,
3. A is a $< \lambda$ -universally Baire set,
4. h is V -generic for $\text{Col}(\omega, < \lambda)$.

Then for some $V[h]$ -generic filter k , there is in $V[h][k]$ an elementary embedding

$$j : L(\mathbb{R}, A)^V \longrightarrow L(\mathbb{R}_h^*, A_h^*)$$

such that $j(A) = A_h^*$.

Proof.

1° Let X be the set of all Woodin cardinals $\delta < \lambda$ which are not limits of Woodin cardinals. By [Kas22, Proposition 3-VII], for some $V[h]$ -generic filter k , there exists in $V[h][k]$ a filter G for $\mathbb{Q}_\lambda$ such that:

- a. for all $\delta \in X$, letting G_δ be $G \cap \mathbb{Q}_\delta^V$, we have that G_δ is V -generic for $\mathbb{Q}_\delta^V$;
- b. letting $i_G^V : V \longrightarrow M$ be the direct limit embedding of the canonical directed system

$$(\text{Ult}(V, G_\delta), i_{G_\delta, G_\epsilon}^V : \delta, \epsilon \in X, \delta \leq \epsilon),$$

we have that κ is in the well-founded part of M ;

- c. $\mathbb{R}^M = \mathbb{R}_h^*$.

Here,

$$i_{G_\delta, G_\epsilon}^V : \text{Ult}(V, G_\delta) \xrightarrow{\Sigma_\omega} \text{Ult}(V, G_\epsilon)$$

is given by $i_{G_\delta, G_\epsilon}^V([f]_{G_\delta}^V) := [f]_{G_\epsilon}^V$.

2° **Claim.** $i_G^V(A) = A_h^*$

Proof. Note that

$$i_G^V(A) = \bigcup_{\delta \in X} i_{G_\delta}^V(A).$$

By Lemma 3, for all $\delta \in X$, $i_{G_\delta}^V(A) = A_{G_\delta}$. This shows that

$$i_G^V(A) = \bigcup_{\delta \in X} A_{G_\delta} = \bigcup_{\delta \in X} (p[T] \cap \mathbb{R}_{G_\delta}) =$$

$$\begin{aligned}
&= p[T] \cap \bigcup_{\delta \in X} \mathbb{R}_{G_\delta} = p[T] \cap \mathbb{R}^M = p[T] \cap \mathbb{R}_h^* = \\
&= \bigcup_{\alpha < \lambda} (p[T] \cap \mathbb{R}_{h \cap \text{Col}(\omega, < \alpha)}) = \bigcup_{\alpha < \lambda} A_{h \cap \text{Col}(\omega, < \alpha)} = A_h^*.
\end{aligned}$$

□

3° Let $j := i_G^V \upharpoonright L_\kappa(\mathbb{R}^V, A)$. It follows that

$$j' : L_\kappa(\mathbb{R}^V, A) \xrightarrow{\Sigma_\omega} L_\kappa(\mathbb{R}_h^*, A_h^*)$$

and $j'(A) = A_h^*$.

4° By [Kas22, Lemma 3-XIV], there exists

$$j : L(\mathbb{R}^V, A) \xrightarrow{\Sigma_\omega} L(\mathbb{R}_h^*, A_h^*)$$

such that $j \supseteq j'$.

5° In particular, $j(A) = A_h^*$, so j is as required.

□

Corollary 5. *Suppose*

1. λ is a limit of Woodin cardinals,
2. $\kappa > \lambda$ is measurable,
3. A is a $< \lambda$ -universally Baire set,
4. g is $< \lambda$ -generic over V .

Then $(\mathbb{R}_g, A_g)^\# \cap V = (\mathbb{R}^V, A)^\#$.

Proof. This is immediate from Proposition 4.

□

Proposition 6. *Suppose*

1. λ is a limit of Woodin cardinals,
2. $\phi(v_x, v_a)$ is a formula,
3. $a \in V$,
4. for every $< \lambda$ -generic filter g over V and every $< \lambda$ -generic filter h over $V[g]$, for all $x \in R_g$,

$$V[g] \models \phi[x, a] \iff V[g][h] \models \phi[x, a],$$

5. for every Woodin cardinal $\delta < \lambda$, it holds in $V^{\mathbb{Q}_\delta}$ that for all $x \in \mathbb{R}$,

$$V[\dot{G}] \models \phi[x, a] \iff \text{Ult}(V, \dot{G}) \models \phi[x, i_G^V(a)],$$

6. A is the set of all $x \in \mathbb{R}$ such that $\phi(x, a)$ holds.

Then A is $<\lambda$ -universally Baire and for every $<\lambda$ -generic filter g over V ,

$$A_g = \{x \in \mathbb{R}_g : V[g] \models \phi[x, a]\}.$$

Proof. See [Kas22, Tree Production Lemma 2-VI]. $\square$

Theorem 7. *Suppose*

1. λ is a limit of Woodin cardinals,
2. $\kappa > \lambda$ is measurable,
3. A is a $<\lambda$ -universally Baire set.

Then

- a. $(\mathbb{R}, A)^\sharp$ is $<\lambda$ -universally Baire,
- b. for every $<\lambda$ -generic filter g over V , $((\mathbb{R}^V, A)^\sharp)_g = (\mathbb{R}_g, A_g)^\sharp$.

Proof.

1° Let T be a $<\lambda$ -absolutely complemented tree that projects onto A , and let $\phi(v_r, v_T)$ be the formula asserting “ $v_r \in (\mathbb{R}, p[v_T])^\sharp$ ”. By Proposition 6, we will be done if we verify that

- a. for every $<\lambda$ -generic filter g over V and every $<\lambda$ -generic filter h over $V[g]$, for all $x \in \mathbb{R}_g$,

$$V[g] \models \phi[x, T] \iff V[g][h] \models \phi[x, T],$$

- b. for every Woodin cardinal $\delta < \lambda$, it holds in $V^{\mathbb{Q}_\delta}$ that for all $x \in \mathbb{R}$,

$$V[\dot{G}] \models \phi[x, T] \iff \text{Ult}(V, \dot{G}) \models \phi[x, i_G^V(T)].$$

2° Let g be $<\lambda$ -generic over V and let h be $<\lambda$ -generic over $V[g]$. We want to show that for all $x \in \mathbb{R}_g$,

$$V[g] \models \phi[x, T] \iff V[g][h] \models \phi[x, T].$$

This comes down to showing that $(\mathbb{R}_{g*h}, A_{g*h})^\sharp \cap V[g] = (\mathbb{R}_g, A_g)^\sharp$, which is immediate from Corollary 5.

3° Now, let $\delta < \lambda$ be a Woodin cardinal, let G be a V -generic filter form $\mathbb{Q}_\delta$, and let us work in $V[G]$. We want to show that for all $x \in \mathbb{R}$,

$$V[G] \models \phi[x, T] \iff \text{Ult}(V, G) \models \phi[x, i_G^V(T)].$$

4° By Lemma 3, $p[i_G^V(T)]^{\text{Ult}(V, G)} = A_G$.

5° So, what we need to show is that $(\mathbb{R}_G, A_G)^\sharp{}^{V[G]} = (\mathbb{R}_G, A_G)^\sharp{}^{\text{Ult}(V, G)}$, which clearly holds.

6° This means that we have verified the conditions to apply Proposition 6, which finishes the proof. $\square$

References

- [Kas22] Obrad Kasum. *Determinacy in $L(\mathbb{R})$* . English. Personal note on the authors webpage, <https://obradkasum.github.io/notes/>. 2022.
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- [Lar04] Paul B. Larson. *The stationary tower. Notes on a course by W. Hugh Woodin*. English. Vol. 32. Univ. Lect. Ser. Providence, RI: American Mathematical Society (AMS), 2004. ISBN: 0-8218-3604-8.

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